

Adjoint-State Method for Stokes Flow

Tags: #CFD #Optimization

In this note, we are going to illustrate the application of adjoint-state method to the equation of Stokes flow.

Equations of Stokes Flow

The equation of stokes flow can be written as a system of reduced order PDE

$$\begin{cases} \nabla^2 \psi(x, y) = \omega(x, y) \\ \nabla^2 \omega(x, y) = 0, \quad \forall (x, y) \in \Omega \end{cases}$$

By imposing the integrals to the first equation, we have

$$\int_{\Omega} \nabla^2 \psi d\Omega = \int_{\partial\Omega} \nabla \psi \cdot \vec{n} d\Gamma = \int_{\Omega} \omega d\Omega$$

The discrete version of this integral can be formulated as

$$\sum^{N_{Cell}} \tilde{\omega} \Delta^2 = \sum^{N_{BC}} [\nabla \psi \cdot \vec{n}] \Delta = A_0$$

where A_0 is a value given by the prescribed Neumann boundary condition of ψ . N_{Cell} is the number of cells of the entire domain, and N_{BC} is the number of cells on the boundary. The Laplace equation of ω is a local constraint which holds in the domain Ω . In the discrete form, this constraint can be formulated as a linear system of equation

$$r(\tilde{\omega}, \omega_{BC}) = A\tilde{\omega} - B = 0$$

To apply the adjoint-state method, let us start with the classical formulation of the problem. We define the optimization problem with the following form:

Optimization Problem

Find the boundary condition ω_{BC} to solve the equation

$$J(\omega, \omega_{BC}) = 0$$

with the constraint $r(\tilde{\omega}, \omega_{BC}) = A\tilde{\omega} - B = 0$.

In this formulation ω_{BC} is called control variable, and the gradient of J is

$$\frac{dJ}{d\omega_{BC}} = \frac{\partial J}{\partial \omega_{BC}} + \frac{\partial J}{\partial \tilde{\omega}} \frac{\partial \tilde{\omega}}{\partial \omega_{BC}}$$

because ω can be considered as a function of its boundary condition. However, this relation can be simplified by an adjoint variable X^* . Let us define a Lagrangian such that

$$\mathcal{L}(\tilde{\omega}, \omega_{BC}, \lambda) = J(\tilde{\omega}, \omega_{BC}) + \lambda r(\tilde{\omega}, \omega_{BC})$$

Thus, the partial derivatives of $\mathcal{L}$ are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \omega_{BC}} &= \frac{\partial J}{\partial \omega_{BC}} + \frac{\partial J}{\partial \tilde{\omega}} \frac{\partial \tilde{\omega}}{\partial \omega_{BC}} + \lambda \left(\frac{\partial r}{\partial \omega_{BC}} + \frac{\partial r}{\partial \tilde{\omega}} \frac{\partial \tilde{\omega}}{\partial \omega_{BC}} \right) \\ \frac{\partial \mathcal{L}}{\partial \tilde{\omega}} &= \frac{\partial J}{\partial \tilde{\omega}} + \lambda \frac{\partial r}{\partial \tilde{\omega}} \end{aligned}$$

The Lagrange multiplier λ can be chosen as

$$\lambda = -\frac{\partial J}{\partial \tilde{\omega}} \left(\frac{\partial r}{\partial \tilde{\omega}} \right)^{-1}$$

to nullify $\frac{\partial \mathcal{L}}{\partial \tilde{\omega}}$. Substituting the λ into the first partial derivative, and defining it as $(X^*)^T$, we get

$$\frac{\partial \mathcal{L}}{\partial \omega_{BC}} = \frac{dJ}{d\omega_{BC}} = \frac{\partial J}{\partial \omega_{BC}} + (X^*)^T \left(\frac{\partial r}{\partial \omega_{BC}} \right)$$

However, if the cost function J is defined as

$$J = \sum_{N_{Cell}} \tilde{\omega} \Delta^2 - A_0$$

we immediately observe that

$$\frac{\partial J}{\partial \omega_{BC}} = 0$$

Thus

$$\frac{dJ}{d\omega_{BC}} = (X^*)^T \left(\frac{\partial r}{\partial \omega_{BC}} \right)$$

In addition, we have

$$dJ = \frac{dJ}{d\omega_{BC}} d\omega_{BC} \approx J^{(n+1)} - J^{(n)} \approx \sum_{j=1}^{N_{BC}} \frac{dJ^{(n)}}{d\omega_{BC_j}^{(n)}} (\omega_{BC_j}^{(n+1)} - \omega_{BC_j}^{(n)})$$

and we want $J^{(n+1)}$ tends to zero as n goes to infinity. However, this is a *underdetermined problem*, since we have only one equation with $N_{BC} > 1$ variables. We can consider the least norm correction by imposing $\omega_{BC}^{(n+1)} = \omega_{BC}^{final}$ and $\omega_{BC}^{(0)}$.

Applying the norm minimization with equality constraints, the resolution of the underdetermined system can be written as the following form:

Minimization with equality constraints

Consider problem

$$\begin{aligned} & \text{minimize} \quad \frac{1}{2} \|\omega_{BC}^{final} - \omega_{BC}^{(0)}\|^2 = \frac{1}{2} \|\delta\omega_{BC}\|^2 \\ & \text{subject to} \quad -J^{(0)} = c^T(\omega_{BC}^{final} - \omega_{BC}^{(0)}) = c^T \delta\omega_{BC} \end{aligned}$$

where c^T is the gradient of the function J .

Let a Lagrangian be defined as

$$\mathcal{L}(\delta\omega_{BC}, \mu) = \frac{1}{2} \|\delta\omega_{BC}\|^2 + \mu(c^T \delta\omega_{BC} + J^{(0)})$$

To reach the minimum, we must have

$$\frac{\partial \mathcal{L}}{\partial \delta\omega_{BC}} = \delta\omega_{BC} + \mu c^T = 0$$

Thus,

$$\delta\omega_{BC} = -\mu c^T$$

Substituting it into the equality constraint, we obtain

$$-\mu \|c\|^2 = -J^{(0)}$$

Therefore,

$$\delta\omega_{BC} = -\frac{J^{(0)}}{\|c\|^2} c^T$$

which implies

$$\omega_{BC}^{final} = \omega_{BC}^{(0)} - \frac{J^{(0)}}{\|c\|^2} c^T$$

Thought

- We must state that this formulation does (intend to) satisfy the total vorticity conservation, which is the priority criterion of the algorithm consistence, but it is unable to obtain the unique solution of the problem. Because, the boundary conditions represents the spatial distribution of the solution on the boundary, thus the spatial information will be lost if we merely consider its integral value. Besides, the Dirichlet condition of ψ is never really used in the criterion of the problem.
- There is also a remark about the vorticity boundary condition ω_{BC} as a control variable. Since the vorticity field depends on the Dirichlet condition and the Neumann Condition of ψ . We have to state that ω_{BC} is not a free variable, and it must belong to some admissible

set, denoted by Σ_{ad} , which can satisfy the given boundary conditions. This should be enforced by an extra constraint.

- At the end of procedure, we observe that the system is underdetermined, thus the solution is not unique unless we can provide extra constraint. (Does such “extra constraint” exist in physics or in mathematics ?)

Reference:

- Andrew M. Bradley, PDE-constrained optimization and the adjoint method
- Gilbert Strang, Computational Science and Engineering, Wellesley-Cambridge Press
- <https://ee263.stanford.edu/lectures/min-norm.pdf>